

Continuum Mechanics

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Chapter 06

Isothermal Fluid Mechanics Part 3



Small Disturbances and Acoustics



Definition (Quiescent State)

Suppose the fluid occupies a fixed region D, there are no body forces, and the initial data are

$$\mathbf{v}_0(\mathbf{x}) = \mathbf{0}, \qquad \rho_0(\mathbf{x}) = \rho^* > 0.$$

The uniform solution

$$\mathbf{v}(\mathbf{x}, t) = \mathbf{0}, \qquad \rho(\mathbf{x}, t) = \rho^*$$

is called a *quiescent state*.

We linearize about this state to describe small-amplitude disturbances.

Small Disturbances

Let $0 \leq \epsilon \ll 1$. Assume

$$|\mathbf{v}_0| = O(\epsilon), \qquad |\rho_0 - \rho^*| = O(\epsilon),$$

and let the boundary have prescribed normal velocity

$$\mathbf{v} \cdot \mathbf{n} = \vartheta = O(\epsilon).$$

We expect

$$|\mathbf{v}| = O(\epsilon), \qquad |\rho - \rho^*| = O(\epsilon).$$

Write the parameter-dependent fields as $\mathbf{v}_\epsilon$ and ρ_ϵ.

Power-Series Expansions



Expand about the quiescent state:

$$\begin{aligned} \mathbf{v}_\epsilon &= \mathbf{0} + \epsilon \mathbf{v}^{(1)} + \epsilon^2 \mathbf{v}^{(2)} + O(\epsilon^3), \\ \rho_\epsilon &= \rho^* + \epsilon \rho^{(1)} + \epsilon^2 \rho^{(2)} + O(\epsilon^3). \end{aligned} \tag{6.32}$$

The coefficients $\mathbf{v}^{(k)}$ and $\rho^{(k)}$ are independent of ϵ. The first-order fields describe the leading disturbance.

Equations Before Collecting Powers



With $\mathbf{b} = \mathbf{0}$, the exact equations are

$$\begin{aligned} \rho_\epsilon \left[\frac{\partial \mathbf{v}_\epsilon}{\partial t} + (\nabla^{\mathrm{x}} \mathbf{v}_\epsilon)\mathbf{v}_\epsilon \right] &= -\pi'(\rho_\epsilon)\nabla^{\mathrm{x}} \rho_\epsilon, \\ \frac{\partial \rho_\epsilon}{\partial t} + \nabla^{\mathrm{x}} \cdot (\rho_\epsilon \mathbf{v}_\epsilon) &= 0. \end{aligned} \tag{6.33}$$

Since $\mathbf{v}_\epsilon = O(\epsilon)$,

$$(\nabla^{\mathrm{x}} \mathbf{v}_\epsilon)\mathbf{v}_\epsilon = O(\epsilon^2).$$

This nonlinear convective term does not enter the first-order equations.

Expansion of Linear Momentum



Substitution of (6.32) into $(6.33)_1$ gives

$$(\rho^* + \epsilon\rho^{(1)} + O(\epsilon^2)) \left[\frac{\partial}{\partial t}(\epsilon \mathbf{v}^{(1)} + O(\epsilon^2)) + \nabla^{\mathrm{x}}(\epsilon \mathbf{v}^{(1)} + O(\epsilon^2))(\epsilon \mathbf{v}^{(1)} + O(\epsilon^2)) \right] \tag{6.34a}$$

$$= -\pi'(\rho^* + O(\epsilon))\nabla^{\mathrm{x}}(\rho^* + \epsilon\rho^{(1)} + O(\epsilon^2)).$$

The terms of order ϵ are

$$\rho^* \frac{\partial \mathbf{v}^{(1)}}{\partial t} = -\pi'(\rho^*)\nabla^{\mathrm{x}}\rho^{(1)}.$$

Expansion of Conservation of Mass



Substitution of (6.32) into $(6.33)_2$ gives

$$\frac{\partial}{\partial t}(\rho^* + \epsilon\rho^{(1)} + O(\epsilon^2)) + \nabla^{\mathrm{x}} \cdot \Big[(\rho^* + \epsilon\rho^{(1)} + O(\epsilon^2))(\epsilon\mathbf{v}^{(1)} + O(\epsilon^2))\Big] = 0. \tag{6.34b}$$

Because ρ^* is constant, the terms of order ϵ are

$$\frac{\partial \rho^{(1)}}{\partial t} + \rho^* \nabla^{\mathrm{x}} \cdot \mathbf{v}^{(1)} = 0.$$

Acoustic Equations

Collecting the first-order terms, including first-order boundary and initial data, gives

$$\left.\begin{aligned} \frac{\partial \mathbf{v}^{(1)}}{\partial t} + \frac{\pi'(\rho^*)}{\rho^*}\nabla^{\mathrm{x}}\rho^{(1)} &= \mathbf{0}, \\ \frac{\partial \rho^{(1)}}{\partial t} + \rho^*\nabla^{\mathrm{x}}\cdot\mathbf{v}^{(1)} &= 0, \\ \mathbf{v}^{(1)}\cdot\mathbf{n} &= \vartheta, \\ \mathbf{v}^{(1)}(\cdot,0) &= \mathbf{v}_0^{(1)}, \\ \rho^{(1)}(\cdot,0) &= \rho_0^{(1)}. \end{aligned}\right\} \tag{6.35}$$

These linear equations describe small deviations from a quiescent elastic fluid.

Proposition (Wave Equation for the Density Disturbance)

The first-order density field satisfies

$$\frac{\partial^2 \rho^{(1)}}{\partial t^2} = \pi'(\rho^*) \Delta^{\mathrm{x}} \rho^{(1)}. \tag{6.36}$$

Proof.

Differentiate conservation of mass in time:

$$\frac{\partial^2 \rho^{(1)}}{\partial t^2} + \rho^* \nabla^{\mathrm{x}} \cdot \frac{\partial \mathbf{v}^{(1)}}{\partial t} = 0.$$

From linear momentum,

$$\frac{\partial \mathbf{v}^{(1)}}{\partial t} = -\frac{\pi'(\rho^*)}{\rho^*} \nabla^{\mathrm{x}} \rho^{(1)}.$$

Proof (Cont.)

Substitution gives

$$\begin{aligned}\frac{\partial^2 \rho^{(1)}}{\partial t^2} &= -\rho^* \nabla^{\mathrm{x}} \cdot \left[-\frac{\pi'(\rho^*)}{\rho^*} \nabla^{\mathrm{x}} \rho^{(1)} \right] \\ &= \pi'(\rho^*) \nabla^{\mathrm{x}} \cdot \nabla^{\mathrm{x}} \rho^{(1)} \\ &= \pi'(\rho^*) \Delta^{\mathrm{x}} \rho^{(1)}.\end{aligned}$$

Because $\pi'(\rho^*) > 0$, the quantity

$$c = \sqrt{\pi'(\rho^*)}$$

is real and is called the *speed of sound* at density ρ^*. □



Example (A Plane Acoustic Wave)

Let $c = \sqrt{\pi'(\rho^*)}$ and choose a unit vector $\mathbf{n}$. For any twice differentiable scalar function A, define

$$\rho^{(1)}(\mathbf{x}, t) = A(\mathbf{n} \cdot \mathbf{x} - ct).$$

Writing $s = \mathbf{n} \cdot \mathbf{x} - ct$, we obtain

$$\frac{\partial^2 \rho^{(1)}}{\partial t^2} = c^2 A''(s), \qquad \Delta^{\times} \rho^{(1)} = |\mathbf{n}|^2 A''(s) = A''(s).$$

Thus $\rho^{(1)}$ satisfies (6.36). The profile travels in the $\mathbf{n}$ direction with speed c without changing shape.

Summary of Sections 6.1 and 6.2



For both ideal and elastic fluids,

$$S = -pI,$$

so the stress is symmetric and supports no shear traction.

For an ideal fluid,

$$\rho = \rho_0, \qquad \nabla^{\chi} \cdot \mathbf{v} = 0,$$

and pressure enforces incompressibility.

For an elastic fluid,

$$p = \pi(\rho), \qquad \pi'(\rho) > 0,$$

and small disturbances satisfy the acoustic wave equation with speed $\sqrt{\pi'(\rho^*)}$.



Newtonian Fluids and the Navier–Stokes Equations



Definition (Incompressible Newtonian Fluid)

A body with reference configuration B is an *incompressible Newtonian fluid* if:

1. Its reference density is uniform: $\rho_0(\mathbf{X}) = \rho_0 > 0$.
2. Its motions are incompressible: $\nabla \cdot \mathbf{v} = 0$.
3. Its Cauchy stress has the form

$$S = -pI + \mathbf{C}(\nabla \mathbf{v}),$$

where p is a scalar pressure field and $\mathbf{C}$ is a constant *fourth-order tensor*. Its action on a second-order tensor A is $[\mathbf{C}(A)]_{ij} = C_{ijkl} A_{kl}$, with summation over k, l.

The fourth-order tensor $\mathbf{C}$ must satisfy the two conditions on the next slide.



Definition (Restrictions on the Viscous Stress)

The *left minor symmetry condition* is

$$[\mathbf{C}(A)]^T = \mathbf{C}(A) \qquad \text{for every } A \in \mathcal{V}^2. \tag{6.37}$$

The *trace condition* is

$$\mathbf{tr}\,\mathbf{C}(A) = 0 \qquad \text{whenever } \mathbf{tr}\,A = 0. \tag{6.38}$$

Thus the viscous stress is symmetric, and it is traceless on incompressible velocity gradients.



Proposition (Density, Symmetry, and Physical Pressure)

For an incompressible Newtonian fluid,

$$\rho(\mathbf{x}, t) = \rho_0, \qquad S^T = S, \qquad p = -\tfrac{1}{3}\,\mathrm{tr}\, S.$$

Consequently, angular momentum balance is automatic, and $-pI$ *and* $\mathbf{C}(\nabla \mathbf{v})$ *are the spherical and deviatoric parts of* S*.*



Proof.

Mass conservation gives $\rho_m \det F = \rho_0$. Incompressibility gives $\det F = 1$, so $\rho_m = \rho_0$ and hence $\rho = \rho_0$.

By (6.37),

$$S^T = -pI + [\mathbf{C}(\nabla \mathbf{v})]^T = -pI + \mathbf{C}(\nabla \mathbf{v}) = S.$$

Since $\operatorname{tr} \nabla \mathbf{v} = \nabla \cdot \mathbf{v} = 0$, (6.38) yields

$$\operatorname{tr} S = -p \operatorname{tr} I + \operatorname{tr} \mathbf{C}(\nabla \mathbf{v}) = -3p + 0.$$

Thus $p = -\frac{1}{3} \operatorname{tr} S$ and the remaining stress is traceless. □

Reactive Stress, Active Stress, and Viscosity



The decomposition

$$S = S^{(r)} + S^{(a)}, \qquad S^{(r)} = -pI, \qquad S^{(a)} = \mathbf{C}(\nabla \mathbf{v})$$

separates the *reactive stress*, enforcing incompressibility, from the *active stress*, specified by the constitutive law.

Result 6.8 will give the frame-indifferent form

$$S^{(a)} = 2\mu L, \qquad L = \mathsf{sym}(\nabla \mathbf{v}) = \tfrac{1}{2}[\nabla \mathbf{v} + (\nabla \mathbf{v})^T].$$

Here L is the *rate of strain tensor* and the constant μ is the *absolute viscosity*. Mechanical passivity requires $\mu \geq 0$.

Proposition (Divergence of the Newtonian Stress)

If μ is constant and $S = -pI + 2\mu\,\mathbf{sym}(\nabla\mathbf{v})$, then

$$\nabla \cdot S = -\nabla p + \mu\Delta\mathbf{v} + \mu\nabla(\nabla\cdot\mathbf{v}).$$

In particular, incompressibility implies

$$\nabla \cdot S = -\nabla p + \mu\Delta\mathbf{v}.$$



Proof.

Use $[\nabla \mathbf{v}]_{ij} = v_{i,j}$ and $[\nabla \cdot S]_i = S_{ij,j}$, with i free and j summed. Then

$$S_{ij} = -p\delta_{ij} + \mu(v_{i,j} + v_{j,i}),$$

so, because μ is constant,

$$\begin{aligned} S_{ij,j} &= -p_{,j}\delta_{ij} + \mu v_{i,jj} + \mu v_{j,ij} \\ &= -p_{,i} + \mu[\Delta \mathbf{v}]_i + \mu(v_{j,j})_{,i}. \end{aligned}$$

Mixed derivatives commute. Since $v_{j,j} = \nabla \cdot \mathbf{v}$, this proves the first formula. The last term vanishes when $\nabla \cdot \mathbf{v} = 0$. □



Proposition (Navier–Stokes Equations)

The balance laws for a Newtonian fluid with constant ρ_0 and μ reduce to

$$\left.\begin{aligned} \rho_0 \left[\frac{\partial \mathbf{v}}{\partial t} + (\nabla \mathbf{v})\mathbf{v} \right] &= \mu \Delta \mathbf{v} - \nabla p + \rho_0 \mathbf{b}, \\ \nabla \cdot \mathbf{v} &= 0. \end{aligned} \right\} \tag{6.39}$$

These are the Navier–Stokes equations. The prescribed field $\mathbf{b}$ is body force per unit mass. The unknowns are $\mathbf{v}$ and p.



Proof.

Linear momentum balance and the material acceleration formula give

$$\rho_0 \dot{\mathbf{v}} = \nabla \cdot S + \rho_0 \mathbf{b}, \qquad \dot{\mathbf{v}} = \frac{\partial \mathbf{v}}{\partial t} + (\nabla \mathbf{v})\mathbf{v}.$$

Insert the preceding stress-divergence identity and use $\nabla \cdot \mathbf{v} = 0$ to obtain $(6.39)_1$.

For constant density, mass conservation becomes

$$\dot{\rho} + \rho \nabla \cdot \mathbf{v} = \rho_0 \nabla \cdot \mathbf{v} = 0.$$

Since $\rho_0 > 0$, this gives $(6.39)_2$. □

Proposition (Pressure Indeterminacy)

If $(\mathbf{v}, p)$ *solves (6.39), then* $(\mathbf{v}, p + f(t))$ *also solves (6.39) for every scalar function* f*.*

Proof.

The spatial derivative of a function of time alone is zero. Thus

$$\nabla[p(\mathbf{x}, t) + f(t)] = \nabla p(\mathbf{x}, t),$$

and every term in (6.39) is unchanged. □

Remark

Pressure is an independent unknown enforcing incompressibility. There is no equation of state $p = \pi(\rho)$ *in this model.*

Example (Shear Traction)

Take $\mathbf{v} = \gamma x_2 \mathbf{e}_1$ in a Cartesian orthonormal frame. Then

$$\nabla \mathbf{v} = \begin{pmatrix} 0 & \gamma & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad 2\mu L = \begin{pmatrix} 0 & \mu\gamma & 0 \\ \mu\gamma & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

On a plane with $\mathbf{n} = \mathbf{e}_2$, the traction is

$$\mathbf{t} = S\mathbf{n} = -p\mathbf{e}_2 + \mu\gamma\mathbf{e}_1.$$

The second term is tangential to the plane. A Newtonian fluid can therefore support shear traction when $\mu\gamma \neq 0$.



Definition (Reynolds Number)

Given a characteristic speed $\vartheta^* > 0$ and length $\ell^* > 0$, the *Reynolds number* for $\mu > 0$ is

$$\mathrm{Re} = \frac{\rho_0 \vartheta^* \ell^*}{\mu}.$$

It compares the scales of convective inertia and viscous force:

$$\frac{\rho_0 (\vartheta^*)^2 / \ell^*}{\mu \vartheta^* / (\ell^*)^2} = \frac{\rho_0 \vartheta^* \ell^*}{\mu}.$$

It is dimensionless. Low Reynolds numbers favor smooth, laminar motion. Large Reynolds numbers permit strong fluctuations and turbulence, but are not by themselves a criterion for turbulence.



Definition (Stokes Equations)

Neglecting the acceleration in (6.39) gives the linear system

$$\mu \Delta \mathbf{v} - \nabla p + \rho_0 \mathbf{b} = \mathbf{0}, \qquad \nabla \cdot \mathbf{v} = 0,$$

called the *Stokes equations*.

Remark

This approximation requires small temporal as well as convective inertia relative to viscous forces. Nearly steady, slow motions with small velocity gradients are the regime described in the text.



Frame Indifference and Mechanical Dissipation

Proposition (Frame Indifference of Incompressibility)

Let $\gamma(F) = \det F - 1$. *Under any superposed rigid motion* $\mathbf{x}^* = Q(t)\mathbf{x} + \mathbf{c}(t)$, *with* $Q(t) \in SO(3)$, *the deformation gradient satisfies* $F^* = QF$ *and the scalar constraint satisfies*

$$\gamma(F^*(\mathbf{X}, t)) = \gamma(F(\mathbf{X}, t)), \qquad \textit{equivalently} \qquad \gamma(QF) = \gamma(F).$$

Thus the value of the constraint is unchanged by a change of observer. In particular, $\gamma(F) = 0$ *if and only if* $\gamma(F^*) = 0$.



Proof.

Under a superposed rigid motion

$$\mathbf{x}^* = Q(t)\mathbf{x} + \mathbf{c}(t), \qquad Q^T Q = I, \qquad \det Q = 1,$$

the deformation gradient transforms as $F^* = QF$. Therefore

$$\gamma(F^*) = \det(QF) - 1 = (\det Q)(\det F) - 1 = \det F - 1 = \gamma(F).$$

Thus $\gamma(F) = 0$ if and only if $\gamma(F^*) = 0$. □

Remark (Frame Indifference of a Constrained Model)

For a constrained constitutive model, it is sufficient that the constraint field and the active stress field be frame-indifferent (Section 6.3.3, following Section 5.6).

For an incompressible Newtonian fluid, under $\mathbf{x}^* = Q(t)\mathbf{x} + \mathbf{c}(t)$*, these two conditions are*

$$\gamma(F^*(\mathbf{X}, t)) = \gamma(F(\mathbf{X}, t)), \qquad \gamma(F) = \det F - 1,$$

$$S^{(a)*}(\mathbf{x}^*, t) = Q(t)S^{(a)}(\mathbf{x}, t)Q(t)^T.$$

We have proved the first condition. Result 6.8 establishes the second.

The reactive stress $-p\mathbf{I}$ *enforces the constraint. Pressure is a constraint multiplier, rather than a separately prescribed constitutive function of the velocity gradient.*

Remark (Frame Indifference of the Full Stress)

Under a superposed rigid motion $\mathbf{x}^* = Q(t)\mathbf{x} + \mathbf{c}(t)$, *we must show*

$$S^*(\mathbf{x}^*, t) = Q(t)S(\mathbf{x}, t)Q(t)^T.$$

Pressure transforms as a scalar: $p^*(\mathbf{x}^*, t) = p(\mathbf{x}, t)$. *Since* $QIQ^T = I$, *the reactive stress already has the required law:*

$$-p^*I = -pI = Q(-pI)Q^T.$$

Thus it remains to show $S^{(a)*}(\mathbf{x}^*, t) = QS^{(a)}(\mathbf{x}, t)Q^T$. *Then, at corresponding points,*

$$S^* = -p^*I + S^{(a)*} = Q(-pI)Q^T + QS^{(a)}Q^T = QSQ^T.$$

Result 6.8 establishes precisely this remaining condition on the active stress. The constraint condition has already been established above.



Result 6.8 (Frame Indifference of Newtonian Active Stress)

The constitutive model $S^{(a)} = \mathbf{C}(\nabla \mathbf{v})$ *is frame-indifferent if and only if, on incompressible motions,*

$$\mathbf{C}(\nabla \mathbf{v}) = 2\mu\, \mathrm{sym}(\nabla \mathbf{v}) = 2\mu L, \tag{6.40}$$

where L *is the rate of strain and* μ *is a scalar constant.*



Proof.

Sufficiency: Choose $\mathbf{C}(A) = 2\mu\,\mathrm{sym}\,A$. First check the constitutive restrictions:

$$[\mathbf{C}(A)]^T = 2\mu(\mathrm{sym}\,A)^T = 2\mu\,\mathrm{sym}\,A = \mathbf{C}(A).$$

Also, $\mathrm{tr}(A^T) = \mathrm{tr}\,A$, so

$$\mathrm{tr}\,\mathbf{C}(A) = 2\mu\,\mathrm{tr}\left(\frac{A + A^T}{2}\right) = 2\mu\,\mathrm{tr}\,A.$$

This is zero whenever $\mathrm{tr}\,A = 0$. Hence (6.37) and (6.38) both hold.

Proof (Cont.)

For $\mathbf{x}^* = Q(t)\mathbf{x} + \mathbf{c}(t)$, the velocity gradient transforms as

$$\nabla^* \mathbf{v}^* = Q(\nabla \mathbf{v})Q^T + \dot{Q}Q^T.$$

Differentiate $QQ^T = I$:

$$\dot{Q}Q^T + Q\dot{Q}^T = 0.$$

Thus $\dot{Q}Q^T$ is skew and contributes no symmetric part. Hence

$$L^* = QLQ^T, \qquad S^{(a)*} = 2\mu L^* = 2\mu QLQ^T = QS^{(a)}Q^T.$$

Proof (Cont.)

Necessity: Write $A = \nabla \mathbf{v}$, with $\mathbf{tr}\, A = 0$. Frame indifference requires

$$\mathbf{C}(QAQ^T + \Omega) = Q\mathbf{C}(A)Q^T, \qquad \Omega = \dot{Q}Q^T.$$

At a chosen instant, take $Q = I$ and prescribe any skew tensor Ω. Setting $A = 0$ and using linearity gives $\mathbf{C}(\Omega) = 0$.

Every trace-free A splits as $A = L + W$, where L is symmetric and trace-free and W is skew. Consequently,

$$\mathbf{C}(A) = \mathbf{C}(L), \qquad \mathbf{C}(QLQ^T) = Q\mathbf{C}(L)Q^T.$$

It remains to determine the action of the fourth-order tensor $\mathbf{C}$ on symmetric, trace-free tensors.

Proof (Cont.)

In an orthonormal frame set $D = \mathbf{diag}(1, -1, 0)$. Rotations by π about each coordinate axis leave D unchanged. The same rotations must leave $\mathbf{C}(D)$ unchanged.

Conjugating by these diagonal rotations changes the signs of the appropriate off-diagonal entries. Thus those entries must vanish:

$$\mathbf{C}(D) = \mathbf{diag}(a, b, c).$$

The proper rotation interchanging $\mathbf{e}_1, \mathbf{e}_2$ and reversing $\mathbf{e}_3$ sends D to $-D$. Equivariance and linearity imply

$$\mathbf{diag}(b, a, c) = \mathbf{C}(-D) = -\mathbf{C}(D) = \mathbf{diag}(-a, -b, -c).$$

Hence $b = -a$, $c = 0$, and $\mathbf{C}(D) = aD$.

Proof (Cont.)

Rotating D shows that $\mathbf{C}(QDQ^T) = aQDQ^T$ for every rotation Q. These rotated copies span the symmetric trace-free tensors:

- Permuting the coordinate axes gives the diagonal differences $\mathbf{e}_i \otimes \mathbf{e}_i - \mathbf{e}_j \otimes \mathbf{e}_j$.
- A $\pi/4$ rotation in the (i, j) plane turns such a difference into $\mathbf{e}_i \otimes \mathbf{e}_j + \mathbf{e}_j \otimes \mathbf{e}_i$.

The two independent diagonal differences and the three off-diagonal symmetric tensors form a basis. By linearity, $\mathbf{C}(L) = aL$ for every symmetric trace-free L.

Writing $a = 2\mu$ gives $\mathbf{C}(A) = 2\mu\,\mathbf{sym}\,A$ whenever $\mathbf{tr}\,A = 0$, proving (6.40). □



Proposition (Viscous Stress Power)

For an incompressible Newtonian fluid,

$$S : L = 2\mu L : L = 2\mu |L|^2.$$

In particular, pressure contributes no stress power. If $\mu \geq 0$, the stress power is nonnegative at every point.



Proof.

Since $S = -pI + 2\mu L$,

$$S : L = -p(I : L) + 2\mu(L : L).$$

But

$$I : L = \mathrm{tr}\, L = \mathrm{tr}(\mathrm{sym}\, \nabla \mathbf{v}) = \mathrm{tr}\, \nabla \mathbf{v} = \nabla \cdot \mathbf{v} = 0.$$

Therefore $S : L = 2\mu L : L$. In components,

$$L : L = L_{ij} L_{ij} \geq 0. \qquad \text{(sum over } i, j\text{)}$$

This establishes both claims. □

Proposition (Mechanical Energy Inequality)

Under the energetic-passivity assumption of Section 5.7, the Newtonian model satisfies the mechanical energy inequality

$$\int_{t_0}^{t_1} \det F(\mathbf{X}, t)\, S_m(\mathbf{X}, t) : L_m(\mathbf{X}, t)\, \mathrm{d}t \geq 0$$

at every material point $\mathbf{X} \in B$*, for every closed isothermal process on every interval* $[t_0, t_1]$*, if and only if* $\mu \geq 0$*.*

Here S_m and L_m are the material descriptions of stress and rate of strain. Incompressibility gives $\det F = 1$.



Proof.

For a closed process on $[t_0, t_1]$, the required inequality is

$$\int_{t_0}^{t_1} (\det F) S_m : L_m \, \mathrm{d}t \geq 0.$$

The preceding proposition gives

$$\int_{t_0}^{t_1} (\det F) S_m : L_m \, \mathrm{d}t = 2\mu \int_{t_0}^{t_1} (\det F) |L_m|^2 \, \mathrm{d}t.$$

Since $\det F = 1$ and $|L_m|^2 \geq 0$, $\mu \geq 0$ is sufficient to guarantee the inequality. □



Initial-Boundary Value Problems

Definition (No-Slip Boundary Condition)

At the interface with a solid having prescribed velocity $\boldsymbol{\vartheta}$, the *no-slip condition* is

$$\mathbf{v} = \boldsymbol{\vartheta}.$$

For a fixed, impermeable solid wall, it becomes $\mathbf{v} = \mathbf{0}$.

With outward unit normal $\mathbf{n}$, decompose

$$\mathbf{v} = (\mathbf{v} \cdot \mathbf{n})\mathbf{n} + \mathbf{v}_{\tan}.$$

At a fixed wall, no slip imposes both $\mathbf{v} \cdot \mathbf{n} = 0$ and $\mathbf{v}_{\tan} = \mathbf{0}$.

Standard Problem in a Fixed Region



Find $\mathbf{v} : D \times [0, T] \to \mathcal{V}$ and $p : D \times [0, T] \to \mathbb{R}$ such that

$$\left.\begin{aligned} \rho_0[\partial_t \mathbf{v} + (\nabla \mathbf{v})\mathbf{v}] &= \mu \Delta \mathbf{v} - \nabla p + \rho_0 \mathbf{b} && \text{in } D, \\ \nabla \cdot \mathbf{v} &= 0 && \text{in } D, \\ \mathbf{v} &= \mathbf{0} && \text{on } \partial D, \\ \mathbf{v}(\cdot, 0) &= \mathbf{v}_0 && \text{in } D. \end{aligned}\right\} \tag{6.41}$$

The first three equations hold for $0 \leq t \leq T$.

The data are the region D, constants $\rho_0 > 0$ and $\mu > 0$, body force $\mathbf{b}$, and initial velocity $\mathbf{v}_0$.



Definition (Compatible Initial Velocity)

Initial data for (6.41) are compatible with incompressibility and the fixed-wall boundary condition if

$$\nabla \cdot \mathbf{v}_0 = 0 \quad \text{in } D, \qquad \mathbf{v}_0 = \mathbf{0} \quad \text{on } \partial D.$$

Remark (Pressure Normalization and Motion)

In a bounded domain with velocity prescribed on the boundary, adding $f(t)$ to p does not change the problem. One may fix this freedom by requiring

$$\int_D p(\mathbf{x}, t)\, \mathrm{d}V_{\mathrm{x}} = 0 \quad \textit{for every } t.$$

Once the Eulerian velocity is known, recover the motion from

$$\partial_t \boldsymbol{\phi}(\mathbf{X}, t) = \mathbf{v}(\boldsymbol{\phi}(\mathbf{X}, t), t), \qquad \boldsymbol{\phi}(\mathbf{X}, 0) = \mathbf{X}.$$

Thus one may determine velocity and pressure before determining particle trajectories.



Remark (Other Domains and Boundaries)

For an ideal or elastic fluid at a fixed impermeable wall, the usual condition is only $\mathbf{v} \cdot \mathbf{n} = 0$. *Tangential slip is allowed.*

For a moving solid, prescribe $\mathbf{v} = \boldsymbol{\vartheta}$ *on its moving interface. Free surfaces and interfaces with other fluids require other conditions. Different parts of the boundary may have different conditions.*

Exterior domains and all of space are also useful. Conditions at infinity control behavior there and may fix the pressure normalization.

Recent Development: Announced Finite-Time Blowup (2026)

OpenAI's September 8, 2026 paper reports the following result. For every $\nu > 0$, there is a smooth, compactly supported force $\mathbf{f}$ on $\mathbb{R}^3 \times (0, \infty)$ and a solution of

$$\partial_t \mathbf{v} + (\nabla \mathbf{v})\mathbf{v} - \nu \Delta \mathbf{v} + \nabla q = \mathbf{f}, \qquad \nabla \cdot \mathbf{v} = 0, \qquad \mathbf{v}(\cdot, 0) = \mathbf{0},$$

Here $\nu = \mu/\rho_0$ and $q = p/\rho_0$. The velocity is smooth for $0 \leq t < 1$, with

$$\sup_{0 \leq t < 1} \|\mathbf{v}(t)\|_{L^2(\mathbb{R}^3)} < \infty, \qquad \limsup_{t \uparrow 1} \|\mathbf{v}(t)\|_{L^\infty(\mathbb{R}^3)} = \infty.$$

Thus the reported result gives failure of global smooth continuation for certain smooth *forced* data, despite bounded kinetic energy. It does not assert blowup for unforced flow.

Source: *Finite Time Blowup for Navier–Stokes*, Theorem 1.1.
CMI's September 11 announcement: evaluation process ongoing.



Kinetic Energy of Fluid Motion



Result 6.9 (Integration by Parts)

Let $\mathbf{v}$ *be a smooth vector field in a regular, bounded region* $D \subset \mathbb{E}^3$*, with* $\mathbf{v} = \mathbf{0}$ *on* ∂D*. Then*

$$\int_D (\Delta \mathbf{v}) \cdot \mathbf{v} \, \mathrm{d}V_x = -\int_D \nabla \mathbf{v} : \nabla \mathbf{v} \, \mathrm{d}V_x.$$



Proof.

Use the summation convention and apply the product rule:

$$\begin{aligned}(\Delta\mathbf{v})\cdot\mathbf{v} &= v_{i,jj}v_i \\ &= (v_{i,j}v_i)_{,j} - v_{i,j}v_{i,j}.\end{aligned}$$

Define $\mathbf{a} = (\nabla\mathbf{v})^T\mathbf{v}$. Its jth component is $a_j = v_{i,j}v_i$. Hence

$$(\Delta\mathbf{v})\cdot\mathbf{v} = \nabla\cdot\mathbf{a} - \nabla\mathbf{v} : \nabla\mathbf{v}.$$

Here i, j are summed indices in the scalar identity; j is free in the formula for a_j.

Proof (Cont.)

Integrating the identity and using the divergence theorem gives

$$\int_D (\Delta \mathbf{v}) \cdot \mathbf{v} \, \mathrm{d}V_\mathrm{x} = \int_D \nabla \cdot \mathbf{a} \, \mathrm{d}V_\mathrm{x} - \int_D \nabla \mathbf{v} : \nabla \mathbf{v} \, \mathrm{d}V_\mathrm{x}$$
$$= \int_{\partial D} \mathbf{a} \cdot \mathbf{n} \, \mathrm{d}A_\mathrm{x} - \int_D \nabla \mathbf{v} : \nabla \mathbf{v} \, \mathrm{d}V_\mathrm{x}.$$

Since $\mathbf{v} = \mathbf{0}$ on ∂D,

$$\mathbf{a} = (\nabla \mathbf{v})^T \mathbf{v} = \mathbf{0} \quad \text{on } \partial D.$$

The boundary integral is zero, which proves the claimed identity. □



Result 6.10 (Poincaré Inequality)

Let D be a regular, bounded region in $\mathbb{E}^3$. There exists a constant $\lambda > 0$, depending only on D, such that

$$\int_D |\mathbf{w}|^2 \, \mathrm{d}V_{\mathrm{x}} \leq \lambda \int_D \nabla \mathbf{w} : \nabla \mathbf{w} \, \mathrm{d}V_{\mathrm{x}}$$

for every smooth vector field $\mathbf{w}$ satisfying $\mathbf{w} = \mathbf{0}$ on ∂D.

The proof below follows the eigenfunction argument in Exercise 6.13.

Proof.

Use the orthonormal Dirichlet eigenfunctions from Exercise 6.13:

$$-\Delta \boldsymbol{\eta}_k = \lambda_k \boldsymbol{\eta}_k, \qquad \boldsymbol{\eta}_k|_{\partial D} = \mathbf{0}, \qquad 0 < \lambda_1 \leq \lambda_2 \leq \cdots,$$

$$\int_D \boldsymbol{\eta}_k \cdot \boldsymbol{\eta}_\ell \, \mathrm{d}V_x = \delta_{k\ell}.$$

Assume the complete eigenfunction expansion, with convergence in the energy norm, as in that exercise:

$$\mathbf{w} = \sum_{k=1}^{\infty} a_k \boldsymbol{\eta}_k.$$

Orthonormality gives $\displaystyle\int_D |\mathbf{w}|^2 \, \mathrm{d}V_x = \sum_{k=1}^{\infty} a_k^2.$

Proof (Cont.)

First take a finite sum $\mathbf{w}_N = \sum_{k=1}^{N} a_k \boldsymbol{\eta}_k$. Result 6.9 and orthonormality imply

$$\begin{aligned}\int_D |\nabla \mathbf{w}_N|^2 \, \mathrm{d}V_\mathrm{x} &= -\int_D \Delta \mathbf{w}_N \cdot \mathbf{w}_N \, \mathrm{d}V_\mathrm{x} \\ &= \sum_{k=1}^{N}\sum_{\ell=1}^{N} \lambda_k a_k a_\ell \delta_{k\ell} = \sum_{k=1}^{N} \lambda_k a_k^2 \\ &\geq \lambda_1 \sum_{k=1}^{N} a_k^2 = \lambda_1 \int_D |\mathbf{w}_N|^2 \, \mathrm{d}V_\mathrm{x}.\end{aligned}$$

Pass to the limit using energy-norm convergence. Taking $\lambda = 1/\lambda_1$ proves the inequality. □



Proposition (Scaling of the Poincaré Constant)

The Poincaré constant has dimensions of length squared. If D is replaced by αD, with $\alpha > 0$, an admissible constant λ is replaced by $\alpha^2 \lambda$.



Proof.

Set $\mathbf{x} = \alpha\mathbf{y}$ and $\mathbf{u}(\mathbf{y}) = \mathbf{w}(\alpha\mathbf{y})$. Then $\mathrm{d}V_\mathrm{x} = \alpha^3\,\mathrm{d}V_\mathrm{y}$ and $\nabla_\mathrm{y}\mathbf{u} = \alpha\nabla_\mathrm{x}\mathbf{w}$. Thus

$$\begin{aligned}\int_{\alpha D} |\mathbf{w}|^2\,\mathrm{d}V_\mathrm{x} &= \alpha^3 \int_D |\mathbf{u}|^2\,\mathrm{d}V_\mathrm{y} \\ &\le \alpha^3\lambda \int_D |\nabla_\mathrm{y}\mathbf{u}|^2\,\mathrm{d}V_\mathrm{y} \\ &= \alpha^2\lambda \int_{\alpha D} |\nabla_\mathrm{x}\mathbf{w}|^2\,\mathrm{d}V_\mathrm{x}.\end{aligned}$$

Also $[\int |\mathbf{w}|^2]/[\int |\nabla\mathbf{w}|^2] = \text{length}^2$. □

Definition (Kinetic Energy)

For a fluid of constant density $\rho_0 > 0$ occupying the fixed region $B_t = D$, its *kinetic energy* at time t is

$$K(t) = \int_{B_t} \tfrac{1}{2}\rho|\mathbf{v}|^2 \, \mathrm{d}V_{\mathrm{x}} = \int_D \tfrac{1}{2}\rho_0|\mathbf{v}|^2 \, \mathrm{d}V_{\mathrm{x}}.$$

Write $K_0 = K(0)$.

We assume a conservative body force $\mathbf{b} = -\nabla\beta$ and introduce

$$\psi = p + \rho_0\beta.$$

Then the pressure and body force combine as $-\nabla p + \rho_0\mathbf{b} = -\nabla\psi$.



Result 6.11 (Kinetic Energy of Newtonian and Ideal Fluids)

Let D be regular and bounded, with fixed boundary, constant density $\rho_0 > 0$, and conservative body force. On every interval of smooth existence:

1. *A Newtonian fluid with $\mu > 0$ and $\mathbf{v} = \mathbf{0}$ on ∂D satisfies*

$$K(t) \leq e^{-2\mu t/(\lambda \rho_0)} K_0.$$

2. *An ideal fluid with $\mathbf{v} \cdot \mathbf{n} = 0$ on ∂D satisfies*

$$K(t) = K_0.$$

Here λ is the constant in Result 6.10.

Proof.

Step 1: Differentiate kinetic energy. The mass-transport identity (Result 5.6) gives

$$\frac{\mathrm{d}}{\mathrm{d}t}K(t) = \frac{\mathrm{d}}{\mathrm{d}t}\int_{B_t} \tfrac{1}{2}\rho|\mathbf{v}|^2 \,\mathrm{d}V_{\mathrm{x}}$$

$$= \int_{B_t} \tfrac{1}{2}\rho\left(|\mathbf{v}|^2\right)^{\cdot} \,\mathrm{d}V_{\mathrm{x}} = \int_{B_t} \rho\dot{\mathbf{v}}\cdot\mathbf{v} \,\mathrm{d}V_{\mathrm{x}}.$$

We used $\left(|\mathbf{v}|^2\right)^{\cdot} = 2\dot{\mathbf{v}}\cdot\mathbf{v}$. Since $B_t = D$ and $\rho = \rho_0$,

$$\frac{\mathrm{d}K}{\mathrm{d}t} = \int_D \rho_0\dot{\mathbf{v}}\cdot\mathbf{v} \,\mathrm{d}V_{\mathrm{x}}. \tag{6.42}$$

Proof (Cont.)

Step 2: Use linear momentum balance. For a Newtonian fluid,

$$\rho_0 \dot{\mathbf{v}} = \mu \Delta \mathbf{v} - \nabla p - \rho_0 \nabla \beta = \mu \Delta \mathbf{v} - \nabla \psi.$$

For an ideal fluid the same identity holds with $\mu = 0$. Substitute into (6.42):

$$\frac{\mathrm{d}K}{\mathrm{d}t} = \int_D [\mu(\Delta \mathbf{v}) \cdot \mathbf{v} - \nabla \psi \cdot \mathbf{v}] \, \mathrm{d}V_{\mathrm{x}}. \tag{6.43}$$

We now evaluate these two terms separately.

Proof (Cont.)

Step 3: Pressure and conservative forcing do no net work. The product rule and incompressibility give

$$\nabla \cdot (\psi \mathbf{v}) = \nabla \psi \cdot \mathbf{v} + \psi \nabla \cdot \mathbf{v} = \nabla \psi \cdot \mathbf{v}.$$

Both boundary conditions imply $\mathbf{v} \cdot \mathbf{n} = 0$. The divergence theorem therefore gives

$$\begin{aligned} \int_D \nabla \psi \cdot \mathbf{v} \, \mathrm{d}V_\mathrm{x} &= \int_D \nabla \cdot (\psi \mathbf{v}) \, \mathrm{d}V_\mathrm{x} \\ &= \int_{\partial D} \psi \mathbf{v} \cdot \mathbf{n} \, \mathrm{d}A_\mathrm{x} = 0. \end{aligned}$$

Only the viscous term in (6.43) remains.

Proof (Cont.)

Step 4: Compute the dissipation. For the Newtonian fluid, $\mathbf{v} = \mathbf{0}$ on ∂D, so Result 6.9 applies to the remaining term:

$$\mu \int_D (\Delta \mathbf{v}) \cdot \mathbf{v} \, \mathrm{d}V_x = -\mu \int_D \nabla \mathbf{v} : \nabla \mathbf{v} \, \mathrm{d}V_x.$$

For the ideal fluid, the viscous term is zero because $\mu = 0$. Thus in either case

$$\frac{\mathrm{d}K}{\mathrm{d}t} = -\mu \int_D \nabla \mathbf{v} : \nabla \mathbf{v} \, \mathrm{d}V_x. \tag{6.44}$$

For the ideal fluid we have not imposed no slip or applied Result 6.9.

Proof (Cont.)

Step 5: Bound the Newtonian dissipation from below. Result 6.10 gives

$$\int_D |\nabla \mathbf{v}|^2 \, \mathrm{d}V_\mathrm{x} \geq \frac{1}{\lambda} \int_D |\mathbf{v}|^2 \, \mathrm{d}V_\mathrm{x}.$$

Multiply by $-\mu < 0$, which reverses the inequality, and use (6.44):

$$\frac{\mathrm{d}K}{\mathrm{d}t} \leq -\frac{\mu}{\lambda} \int_D |\mathbf{v}|^2 \, \mathrm{d}V_\mathrm{x} = -\frac{2\mu}{\lambda \rho_0} K(t). \tag{6.45}$$

The last equality follows from $\int_D |\mathbf{v}|^2 \, \mathrm{d}V_\mathrm{x} = 2K(t)/\rho_0$.

Proof (Cont.)

Step 6: Integrate the differential inequality. Set $a = 2\mu/(\lambda\rho_0) > 0$. Equation (6.45) gives

$$K'(t) + aK(t) \leq 0.$$

Multiply by e^{at} and use the product rule:

$$\frac{\mathrm{d}}{\mathrm{d}t}[e^{at}K(t)] = e^{at}[K'(t) + aK(t)] \leq 0.$$

Integrating from 0 to t yields

$$e^{at}K(t) - K_0 \leq 0, \qquad K(t) \leq e^{-at}K_0 = e^{-2\mu t/(\lambda\rho_0)}K_0.$$

For the ideal fluid, (6.44) gives $K'(t) = 0$, hence $K(t) = K_0$. □

Proposition (Consistency with the Local Stress Power)

For a divergence-free velocity with $\mathbf{v} = \mathbf{0}$ *on* ∂D,

$$2 \int_D L : L \, \mathrm{d}V_{\mathrm{x}} = \int_D \nabla \mathbf{v} : \nabla \mathbf{v} \, \mathrm{d}V_{\mathrm{x}}.$$

Thus the global dissipation in (6.44) equals $\int_D 2\mu L : L \, \mathrm{d}V_{\mathrm{x}}$.



Proof.

Let $A = \nabla \mathbf{v}$ and $L = (A + A^T)/2$. Expanding gives

$$2L : L = A : A + A : A^T.$$

The cross term has integral zero. Indeed, summing i, j,

$$\begin{aligned}\int_D v_{i,j} v_{j,i} \, \mathrm{d}V_\mathrm{x} &= \int_{\partial D} v_i v_{j,i} n_j \, \mathrm{d}A_\mathrm{x} - \int_D v_i v_{j,ij} \, \mathrm{d}V_\mathrm{x} \\ &= 0 - \int_D v_i (v_{j,j})_{,i} \, \mathrm{d}V_\mathrm{x} = 0.\end{aligned}$$

The boundary term vanishes by no slip, and the volume term vanishes by incompressibility. Integrating $2L : L = A : A + A : A^T$ proves the claim. □



Definition (Kinematic Viscosity and Decay Time)

Define the kinematic viscosity $\nu = \mu/\rho_0$. The energy estimate is

$$K(t) \leq e^{-2\nu t/\lambda} K_0.$$

The energy-decay time scale is $\lambda/(2\nu)$. Larger viscosity relative to density gives faster decay.

For a fixed shape enlarged by a factor α, the Poincaré constant scales as α^2. Thus the decay time scale grows by α^2.

Remark (Meaning and Scope of the Energy Estimate)

An ideal fluid has no viscous mechanism for dissipating kinetic energy. A Newtonian fluid with $\mu > 0$ dissipates energy through viscous stress.

The exponential bound uses a bounded region, fixed walls with no slip, and a conservative body force. Moving walls or nonconservative forcing can supply energy to the fluid.

If a smooth Newtonian solution exists for all $t \geq 0$ under these assumptions, then $K(t) \to 0$ exponentially. When $\mu = 0$, the bound reduces to $K(t) \leq K_0$ and does not imply decay.